Leakage-Aware Deep Learning for Financial Event-State Forecasting

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ABSTRACT

This project studies whether aligned company-news event states improve multi-horizon excess-return forecasting beyond strong market baselines. I built a local, reproducible artifact pipeline for the top-50 liquid U.S. equities, with same-day-as-close news alignment, no-news states, purged expanding folds, an untouched lockbox, fold-local preprocessing, saved predictions, tuning ledgers, and conservative claim gates. The strongest current active model is an updated Alpha Vantage XGBoost model: 800 development trials produced a post-selection lockbox weighted error delta of -5.34×10^{-4} versus the prior XGBoost baseline. A no-direct-TSMixer horizon mixer improved validation error, but degraded versus XGBoost on lockbox, illustrating selection pressure. A historical portfolio simulation is reported only as a diagnostic. The main conclusion is methodological: in noisy financial forecasting, leakage control and non-claiming uncertainty analysis are as important as model complexity.

1 INTRODUCTION

Financial news may contain useful short-lived information, but return prediction is low signal-to-noise and highly vulnerable to leakage, multiple testing, and post-hoc trading narratives. The project therefore frames the task as end-of-day forecasting of forward excess returns for horizons $\{1, 5, 10, 20, 60\}$ days, not as price prediction or trading advice. The contribution is a reproducible deep-learning-ready benchmark: top-50 equities, Alpha Vantage news, Twelve Data OHLCV, explicit no-news rows, purged walk-forward development folds, a final lockbox, and artifact-backed model comparisons. The implemented roster spans linear and tree market baselines, FinBERT feature sidecars, TSMixer sequence diagnostics, XGBoost tuning, and horizon-level XGB/TSMixer mixing. AUC is intentionally not reported: the target is continuous excess return, so RMSE/MAE, zero-baseline R^2, rank IC, bootstrap support, and lockbox deltas are the relevant metrics.

2 RELATED WORK

The design follows market-efficiency and asset-pricing cautions from Fama, Fama-French, and Campbell-Lo-MacKinlay [4, 7, 8]: small apparent edges require strict out-of-sample controls. Textual-finance work shows why news can matter but also why representation matters [15, 17]. Modern NLP motivates contextual encoders and FinBERT-style finance adaptation [2, 6]. For tabular market features, gradient boosting remains a strong baseline [5, 12]. For sequence modeling, the relevant state of the art includes interpretable multi-horizon attention, patch-based transformers, inverted transformers, and foundation time-series models [1, 13, 14, 16]. Evaluation follows forecasting-metric and time-series validation guidance [3, 11]; multiple-testing caution follows Harvey et al. [10]. ROC-AUC is cited only to justify exclusion for this regression target [9].

Table 1: Replicable result summary. Negative deltas mean lower error.

Item	Split/metric	Value
Stage 43 XGB	trials / best trial	800 / 798
Stage 43 XGB	lockbox delta vs prior XGB	-5.34×10^{-4}
Stage 43 XGB	lockbox both-metric improvement	62.2%
Stage 48 mixer	trials / best trial	800 / 391
Stage 48 mixer	validation delta vs XGB	-2.97×10^{-4}
Stage 48 mixer	lockbox delta vs XGB	$+9.20 \times 10^{-5}$
Stage 48 mixer	lockbox improvement vs XGB	11.1%
Portfolio diagnostic	rows / initial capital	75 / $100k
Best portfolio row	final equity / return	$115,045 / 15.0%
ROC-AUC	applicability	N/A: regression target

3 EXPERIMENTS

Data and splits. The benchmark uses 50 liquid equities, five horizons, forward leave-one-out excess-return labels, and explicit event-state rows for no-news dates. The enhanced Alpha Vantage rebuild aligned 166,711 news rows while preserving provider-truncation limitations. Stage 4 defines nine expanding walk-forward development folds plus a held-out lockbox ending 2026-02-13; lockbox rows are never used for tuning.

Models and training. The main tuned model is `updated_av_xgboost_event_` selected from 800 Optuna trials by a guarded validation objective over RMSE/MAE horizon deltas. The TSMixer branch exposed a readout bug: prior mean-time pooling was stale; the corrected readout is flattened time-feature state. Because a corrected full TSMixer search was estimated at 23.1 hours and was rejected, Stage 48 tuned only a lightweight horizon mixer over frozen Stage 43 XGBoost and frozen Stage 44 TSMixer predictions; no TSMixer weights were retrained.

Results. Figure 1 and Table 1 show the key pattern. XGBoost improved against the prior XGBoost family on lockbox, but the Stage 48 mixer is a cautionary counterexample: it improved validation and the frozen TSMixer diagnostic input, yet was worse than XGBoost on lockbox. Stage 25 block-bootstrap evidence was supportive for the earlier frozen XGBoost candidate, but final claim gates remain conservative.

4 DISCUSSION

The strongest lesson is that financial deep learning must be evaluated as a system, not as a model leaderboard. The project found real engineering gains: leakage-aware splits, saved predictions, uncertainty diagnostics, tuning traces, and stronger market baselines. It also found failure modes: stale TSMixer readout semantics, validation-to-lockbox degradation, and portfolio-like diagnostics that can look

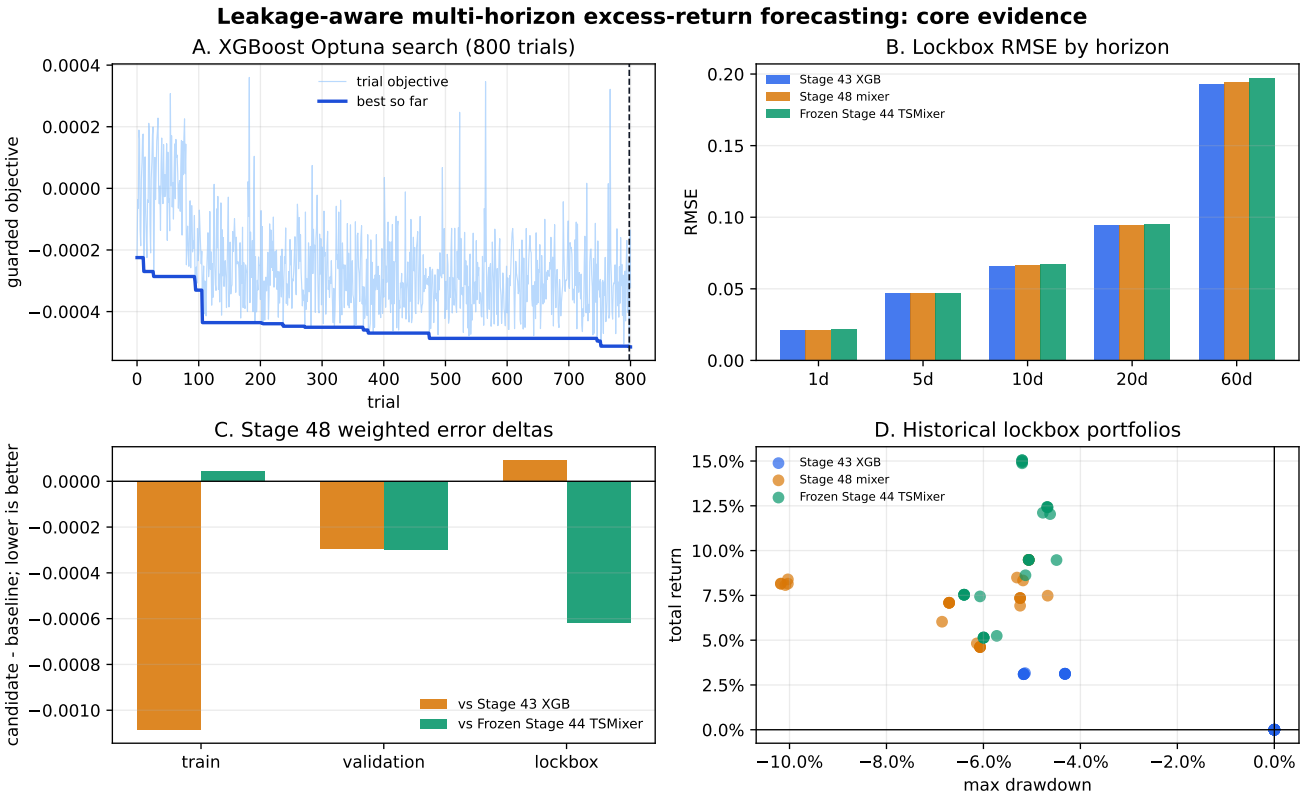


Figure 1: Core evidence. A: XGBoost tuning converged over 800 train/validation-only trials. B: lockbox RMSE by horizon. C: Stage 48 validation gains over XGBoost did not transfer to lockbox. D: the portfolio simulation is diagnostic only; high returns are not claim evidence.

persuasive while remaining non-causal and assumption-dependent. The transparent conclusion is therefore narrow: the updated XGBoost pipeline is the current strongest active evidence; the frozen-TSMixer mixer did not earn a positive lockbox claim; event-state value is not proven; and all outputs remain historical diagnostics, not live forecasts or trading signals.

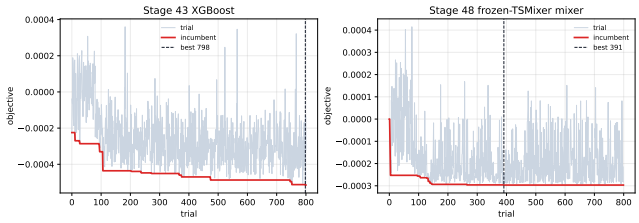


Figure 2: Optuna incumbent curves for Stage 43 XGBoost and Stage 48 horizon mixer.

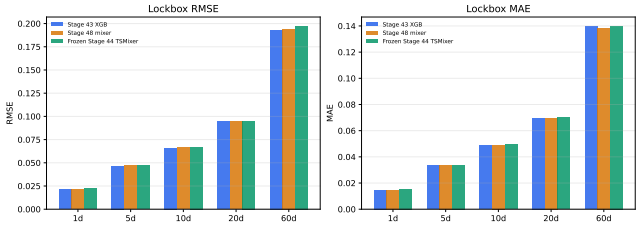


Figure 3: Lockbox RMSE and MAE by horizon for XGBoost, frozen TSMixer diagnostic input, and Stage 48 mixer.

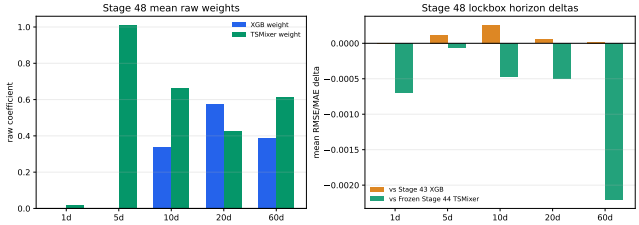


Figure 4: Stage 48 horizon-specific raw mixer weights and lockbox horizon deltas.

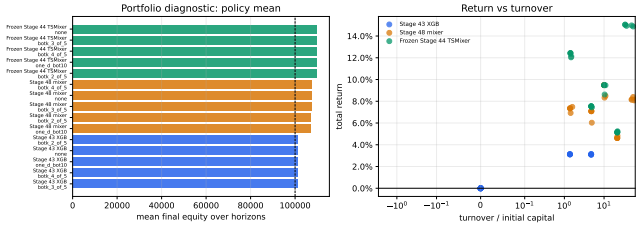


Figure 5: Historical portfolio diagnostics: final equity by policy and return versus turnover.

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A ARTIFACT ROOTS AND REPLICATION COMMANDS

Primary evidence roots are: Stage 43 XGBoost tuning `data/v1p0_full_results/`
Stage 48 horizon mixer `...stage48__frozen_tsmixer_horizon_mixer__20260`
Stage 47 portfolio simulation `...stage47__lockbox_portfolio_simulation__`
Validation command: `source ~/venvs/ai/bin/activate && python -m pytest tests`.

B SUPPLEMENTARY VISUALIZATIONS